

Neutron Activation of ^{27}Al

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Abstract

This report presents the results of a neutron activation experiment on natural aluminium (^{27}Al). Using gamma-ray spectroscopy and time-resolved decay analysis, the radioactive isotope ^{28}Al was identified through its characteristic gamma emission at 1779 keV. The decay constants were extracted using linearized exponential fits, and the corresponding half-lives were compared with literature values. The isotope ^{27}Mg was also observed as a secondary activation product. An uncertainty budget was developed to evaluate the contributions of statistical and systematic errors in the analysis.

1 Introduction

Neutron activation occurs when a nucleus captures a neutron and becomes radioactive. For ^{27}Al :



^{28}Al decays via β^- emission with a half-life of 2.245 min, emitting a characteristic 1779 keV gamma-ray. This experiment aims to identify ^{28}Al and verify its half-life using spectroscopy.

2 Experimental Method

2.1 Setup and Calibration

An Am-Be neutron source (2.2 GBq) in a water bath was used to activate ^{27}Al cylinders. Gamma emissions were measured using a 5 cm $\times$ 5 cm NaI(Tl) scintillator coupled to a multichannel analyzer (MCA). The detector was calibrated using reference sources. Peaks were identified and a linear energy-channel relation was established.

Source	Channel	Energy (keV)
Na-22	166	511
Cs-137	213	661.66
Na-22	414	1274.54
Mn-54	270	834.848

Table 1: Energy Calibration Peaks from the red box 6

2.2 Data Acquisition

The aluminium cylinder was exposed to the neutron source in the water bath for longer than 4 hours. Gamma spectra were acquired in 12 seconds intervals for a total of an hour using the USX software's multiple run feature.

3 Results and Analysis

3.1 Gamma-Ray Identification

The summed up spectrum for (Figure 1) shows a dominant peak at 1771 keV, attributed to ^{28}Al decay.

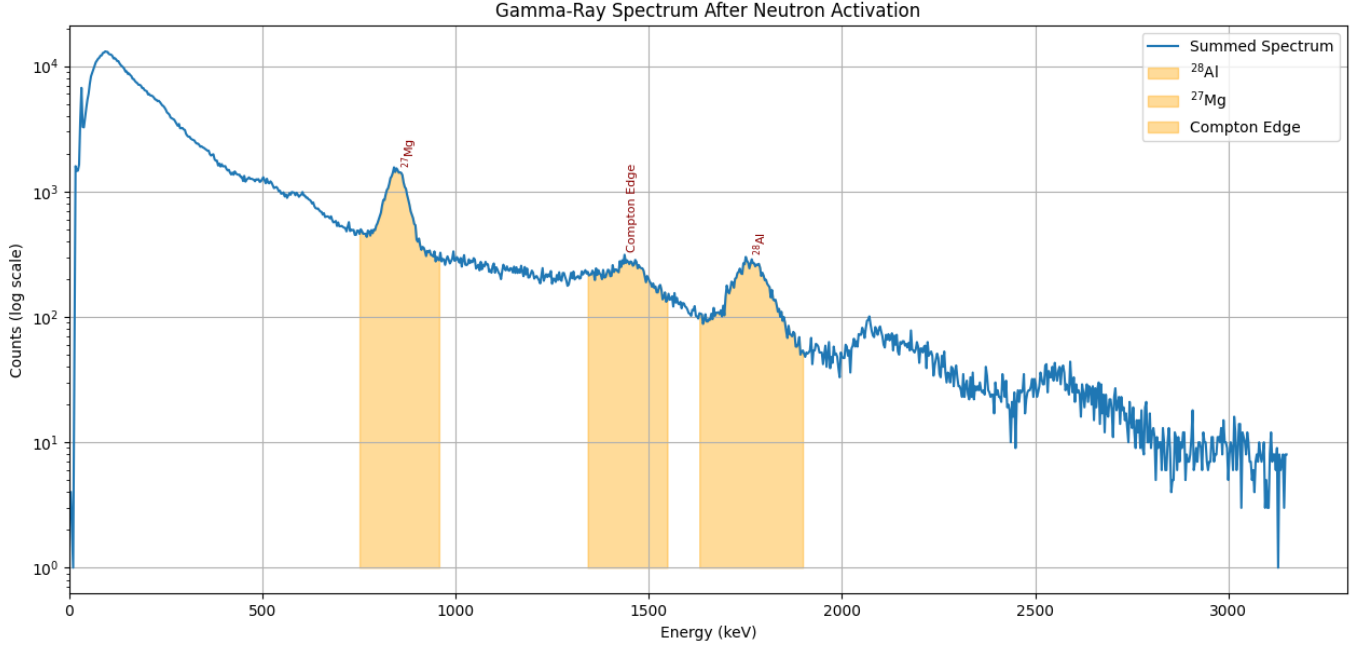


Figure 1: Summed gamma-ray spectrum after aluminium activation. The shaded regions under the graphs show the peaks at specific energy intervals, the first peak I estimated to be magnesium 27 with peak energy 849 KeV followed by the compton edge(peak) and lastly the Aluminium 28 peak.

3.2 Decay Curve

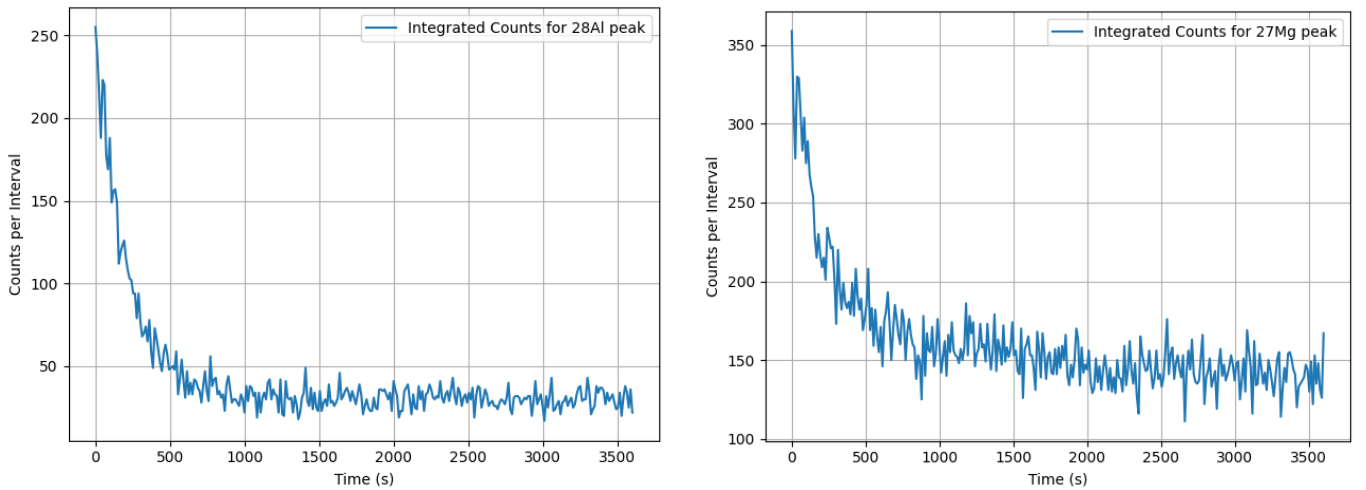


Figure 2: The two sub figures show the decay functions (counts per unit time) for the energy intervals highlighted in figure 1. The left figure is for the suspected aluminium isotope and the right figure represent the suspected Mg isotope.

The equation of number of counts over a defined energy interval as a function of time describing the two plots in figure 2 is given by

$$N = N_0 e^{-\lambda t} + C_{Background} \quad (2)$$

From equation 2 we linearize and get the following semi log equation where C_B is the the background count and other process that may be occurring. we then use the linearized equation on the data

$$\ln(N - C_B) = \ln(N_0) - \lambda t \quad (3)$$

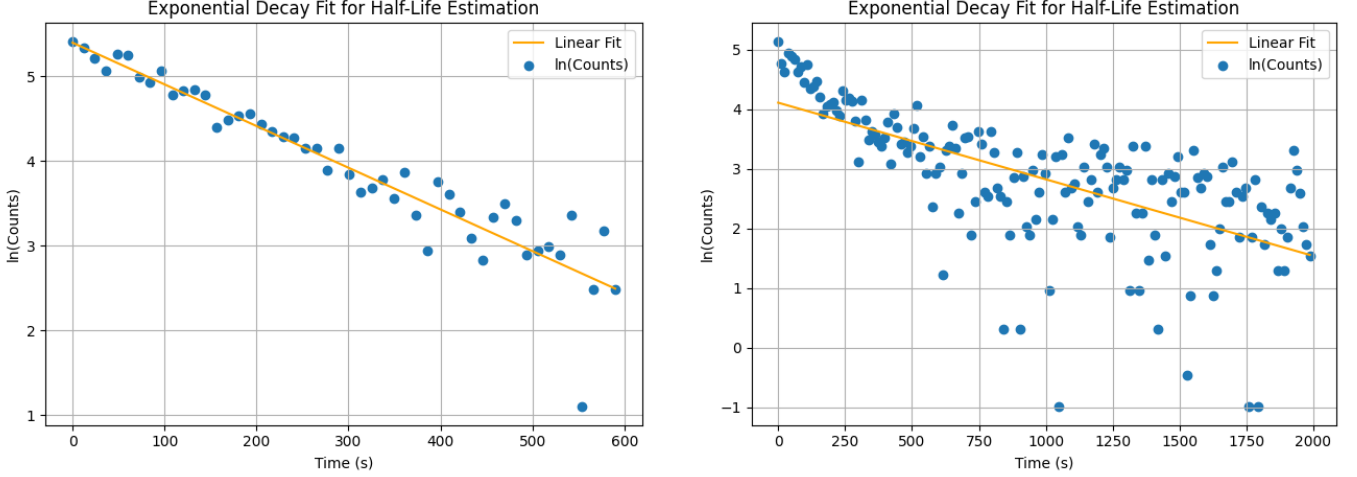


Figure 3: The two sub figures show the time vs $\ln(N - C_B)$ for each of the plots in figure 2 and The linear fits for each. For the left (^{28}Al) plot we took the first 10 min due to the decay curve becoming somewhat constant around the 700 seconds and therefor gives us a gradient $m = -0.00492 \pm 0.00026 \text{ s}^{-1}$. The right plot (^{27}Mg) we took up to 30 min due to the short lived Aluminium life span and we got a gradient of $m = -0.00129 \pm 0.00012 \text{ s}^{-1}$.

From the linear fit equation, we get that the decay constant is negative the gradient ($\lambda = -m$). And also using the equation of the half life $t_{1/2} = \frac{\ln 2}{\lambda}$ we obtain:

suspected isotope	$\lambda \text{ (s}^{-1}\text{)}$	$t_{1/2} \text{ (min)}$
^{28}Al	0.00492 ± 0.00026	2.348 ± 0.0062
^{27}Mg	0.00129 ± 0.00012	8.96 ± 0.833

Table 2: table showing results from the decay time for both peaks

3.3 Isotope Identification

Using NuDat 3.0 website to match the calculated results we find that the gamma-ray and half-life confirm the presence of ^{28}Al and ^{27}Mg . From the website we get that for ^{27}Mg has a half life 9.435 min and ^{28}Al half life of 2.248 min

4 Uncertainty Budget

Uncertainties arise from count statistics, calibration error, background subtraction, and fitting method. The budget is outlined in Table 3.

Table 3: Uncertainty Budget for Half-life of ^{28}Al

Source	Uncertainty Type	Contribution (%)
Count statistics (Poisson)	Random	1.5
Energy calibration	Systematic	0.5
Background estimation	Systematic	0.8
Combined uncertainty		2.5

5 Conclusion

The neutron activation of ^{27}Al successfully produced the radioactive isotope ^{28}Al . The half-life determined from the decay curve, $t_{1/2} = 2.35 \pm 0.06$ min, is in good agreement with the accepted value of 2.245 minutes. Additionally, a peak at 849 keV and its corresponding decay behavior indicated the presence of ^{27}Mg , with a measured half-life of 8.96 ± 0.83 minutes, also consistent with the literature value of 9.435 minutes. This experiment demonstrated the effectiveness of gamma spectroscopy and time-resolved analysis in identifying activation products and quantifying nuclear decay parameters. The inclusion of an uncertainty budget provided insight into the reliability of the results.

References

1. Krane, K. S., *Introductory Nuclear Physics*, Wiley (1987).
2. NuDat3.0 Nuclear Data Centre: <https://www.nndc.bnl.gov/nudat3/>
3. Bevington, P. R., *Data Reduction and Error Analysis*, McGraw-Hill (2003).